Machine Learning in Asset Pricing

MFE5340: AI in FE: Quantitative Investment

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- Financial Data
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- Feature Selection and Labeling
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- Facts and Fictions



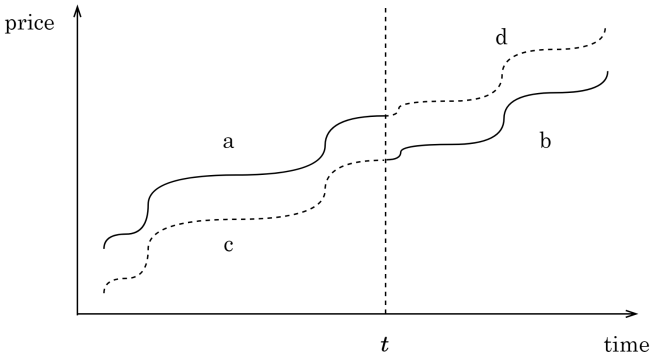
Price and Volume Data

Price and volume data primarily refers to market-related data, including

- open price,
- high price,
- low price,
- close price,
- trading volume,
- turnover rate,
- limit-up or limit-down information.

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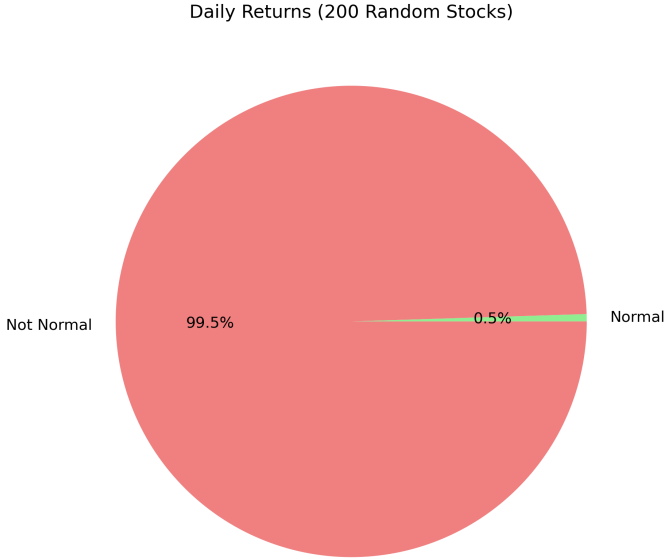
- Events such as stock dividends, splits, and cash dividends can create large gaps in stock prices. To maintain data continuity, forward and backward adjustments are commonly used.
- **Forward adjustment** maintains the current price while adjusting historical prices.
- **Backward adjustment** ensures that historical prices remain unchanged, and adjusts current price.

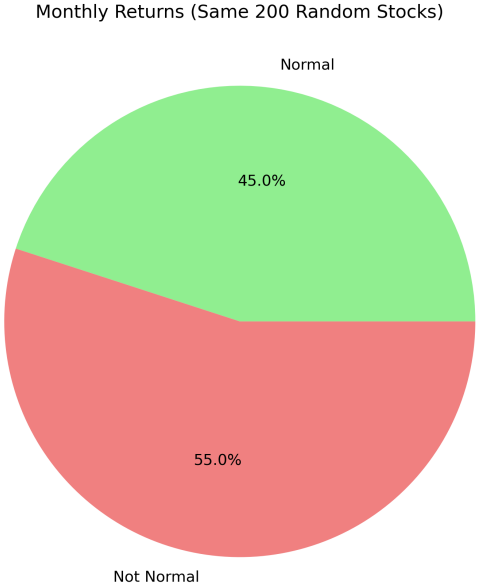


no adjustment: a + b
backward adjustment: a + d
forward adjustment: c + b

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Returns Are Not Normal

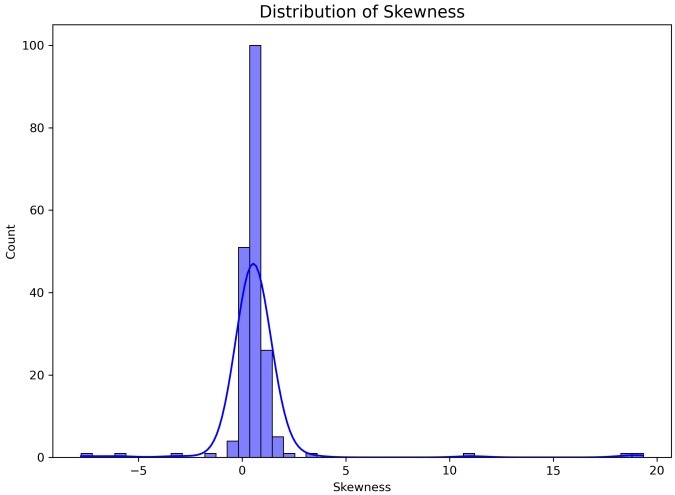


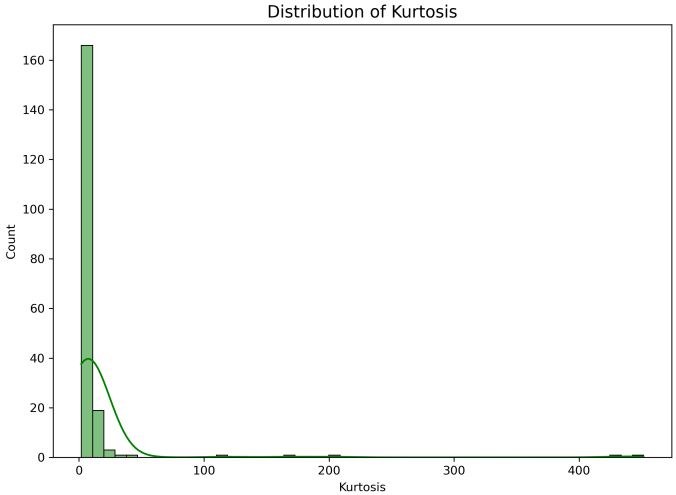


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Returns Are Not Normal

1 Financial Data





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Untradable Stocks: Suspended, Limit-Up, and Limit-Down

On rebalancing days, the following stocks are excluded from trading:

- **Suspended Stocks**: These stocks are excluded because they cannot be traded on the market.
- **Limit-Up Stocks**: Stocks that have reached the upper price limit and cannot be traded are also excluded.
- **Limit-Down Stocks**: This is to restricting the sale of limit-down stocks in short portfolios, making factor returns more consistent with real-world scenarios.

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Financial Data Reporting

- **Financial Data** refers to data related to a company's financials, reflecting the results of operations and changes in financial conditions. Stock exchanges require listed companies to regularly publish financial reports to protect investors' interests.

- **Reporting Period**: The fiscal year in the Chinese stock market starts on January 1 and ends on December 31. Listed companies are required to disclose financial reports four times a year:
  - **First-quarter report:** Covering the period from January 1 to March 31.
  - **Half-year report (i.e., interim report):** Covering the period from January 1 to June 30, including comprehensive data for both the first and second quarters. (There is no separate second-quarter report.)
  - **Third-quarter report:** Covering the period from January 1 to September 30.
  - **Annual report:** Reflects the financial situation for the entire year, covering the period from January 1 to December 31. (There is no separate fourth-quarter report.)

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Financial Reporting Period Table

Report	Start Date	End Date	Latest Disclosure Date
Q1 Report	January 1	March 31	April 30 (same year)
Half-year Report	January 1	June 30	August 31 (same year)
Q3 Report	January 1	September 30	October 31 (same year)
Annual Report	January 1	December 31	April 30 (next year)

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Three Financial Statements

Statement	Definition	Type
Balance Sheet	Reflects the company's assets, liabilities, and owner's equity as of a specific date (typically the end of the accounting period). It is a **snapshot** of the financial position.	A specific date
Income Statement	Shows the company's operational results (profit or loss) over a certain period.	Interval data
Cash Flow Statement	Highlights how different components of the balance sheet affect cash flow, categorized into operations, investing, and financing activities.	Interval data

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Quarterly Data

- For **Balance Sheet**, since it reports financial results for a specific date (i.e., end of accounting period), the single-quarter data is the same as the corresponding reporting period data.
- For **Income Statement** and **Cash Flow Statement**, Q1 data is the same as the first-quarter report, and data for the remaining three quarters is the difference between the current period and the previous period.

Quarter	Balance Sheet	Income Statement	Cash Flow Statement
Q1	Q1 Report	Q1 Report	Q1 Report
Q2	Half-year Report	Half-year Report − Q1 Report	Half-year Report − Q1 Report
Q3	Q3 Report	Q3 Report − Half-year Report	Q3 Report − Half-year Report
Q4	Annual Report	Annual Report − Q3 Report	Annual Report − Q3 Report

- If prior period data is missing, single-quarter data cannot be calculated and will be empty.

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Trailing Twelve Months (TTM) Data

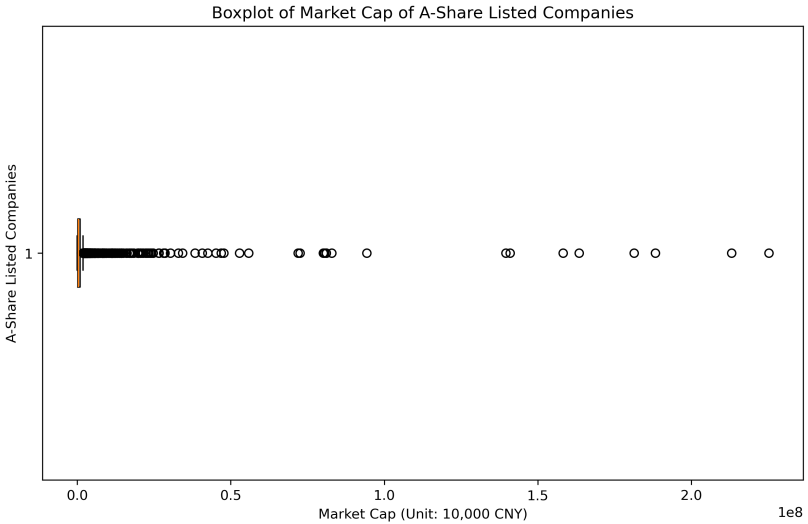
- **Trailing Twelve Months (TTM)** represents data for the past 12 months, used to eliminate seasonality and improve comparability across periods.

- A key consideration is how to handle **balance sheet** items (technically, not quite TTM). There are two common methods:
  1. Use the average of the most recent four reporting periods.
  2. Use the average of the current latest report and the corresponding report from the previous year.

- For income statement and cash flow statement:
  - If the latest reporting period is the annual report, TTM data equals the annual report data.
  - If it is any other report, TTM data equals the current latest report plus last year's annual report, minus the corresponding period from last year.

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Example: XS Distribution

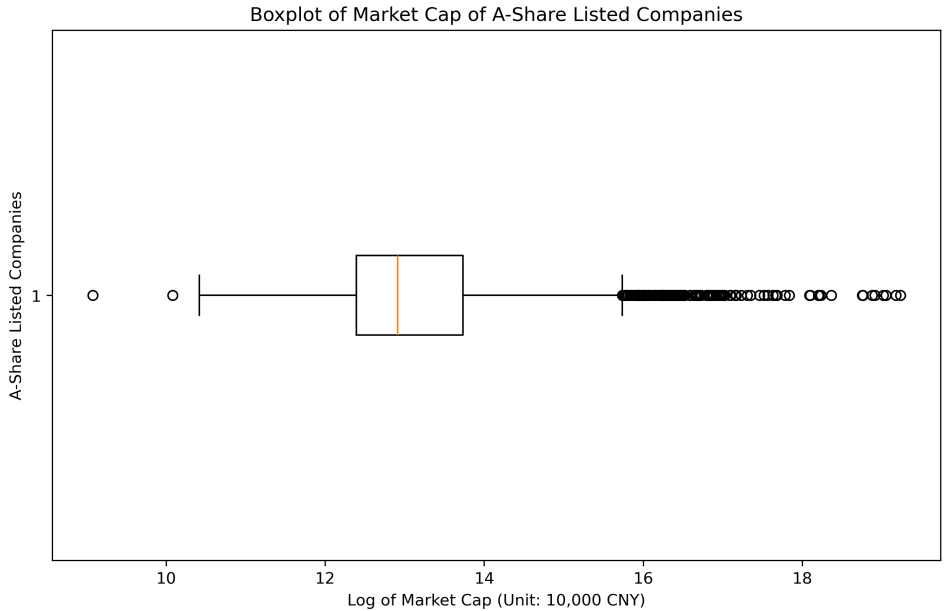


- Total market capitalization as of 2024/8/31.

Statistic	Value (100M)
Mean	159.2
Std	830.3
Min	0.88
25%	24.2
50%	40.5
75%	92.0
Max	22,520.2

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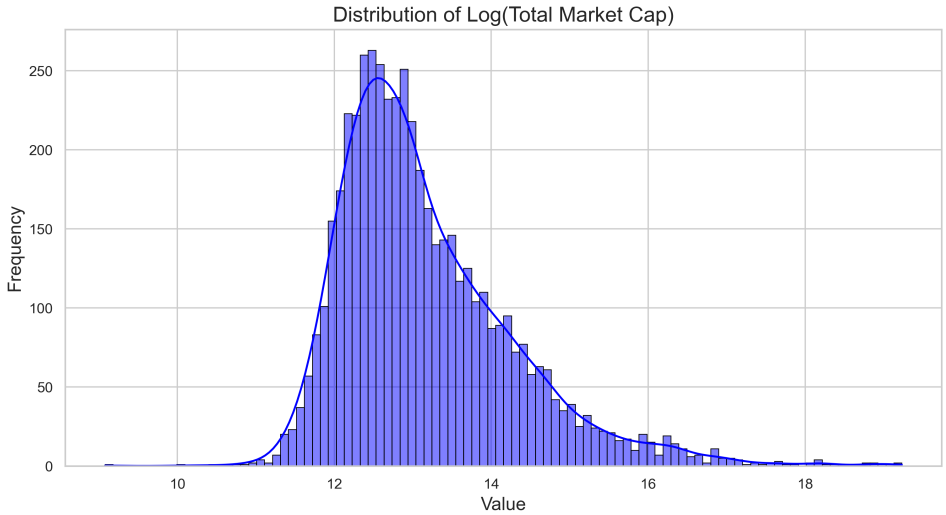
Example: XS Distribution



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Example: XS Distribution

- Log of Total Market Cap:



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Stock Pool/Universe

- Special treatment of the financial sector: Companies in the financial sector (e.g., banking, securities, and insurance) are typically excluded from empirical studies due to significant differences in their business models and accounting items compared to regular firms. This has become a standard practice in factor investing.

- But it depends:
  - Most of the factors proposed are universal and are not affected by industry-specific accounting items.
  - The financial sector includes many high-quality companies, which are beneficial from an investment perspective.
  - In classic academic studies, such as those by Fama and French, financial stocks were not deliberately excluded.

- **Be cautious about data snooping bias!**

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Exclusion of Blacklist Stocks

Certain stocks are considered to have poor financial quality and high risk. Including these stocks in the stock pool may distort the evaluation of factor returns. These stocks are referred to as the blacklist:

- **Delisting warning stocks (*ST):** A company has reported losses for three consecutive years.
- **Risk warning stocks (ST):** Stocks flagged due to two consecutive years of negative net income or a last-year audited net asset value below par value. These stocks face suspension or delisting risks.
- **Newly listed stocks:** Defined as stocks listed for less than one year. These stocks tend to exhibit high volatility and are often subject to speculative trading.
- **Stocks with negative net assets**.

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Handeling Outliers

- A larger variance of the predictor variable in the cross-section implies a greater difference in information. However, the variable should fall within a reasonable range.
- Due to issues such as data quality or calculation methods, outliers may arise, leading to interference in subsequent analysis.

> For example, on May 28, 2019, the ROE distribution showed a minimum value of -1227.85%, which contradicts common sense. This outlier was traced back to **excessive losses** and **very small net assets** for 600961.SH.

Stock Count	Mean	Std. Dev	Min	25%	Median	75%	Max
3365	4.16%	34.53%	−1227.85%	2.87%	7.03%	11.69%	78.51%

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Handeling Outliers

- **Winsorization**: the most common method, widely used in academia and industry. It **removes** or **replaces** values below the pth percentile and above the $(1-p)^{th}$ percentile. (Common choice of p is 1.)

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- **3-Sigma Rule**: Values outside of three standard deviations from the mean are outliers. But outliers can influence the calculation of the mean and standard deviation, leading to inaccurate results.

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- **3-Sigma Rule**: Values outside of three standard deviations from the mean are outliers. But outliers can influence the calculation of the mean and standard deviation, leading to inaccurate results.
- **Median Absolute Deviation**: it replaces the mean with the median and the standard deviation with the MAD:

$$\text{MAD} = \text{median}(|X_i - X^m|)$$

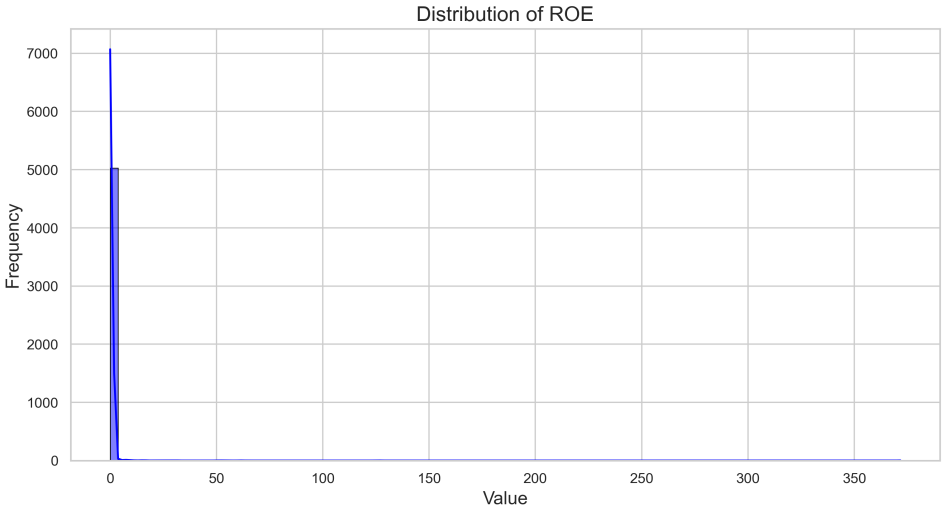
where X^m is the median. The threshold is set to $3 \times \text{MAD}_e$, where $\text{MAD}_e = 1.4826 \times \text{MAD}$ (Why?).

Question for you: Are these best industry practices? Can you think of other, yet better, methods to handle outliers?

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Example: Impact of Outliers on XS Distribution

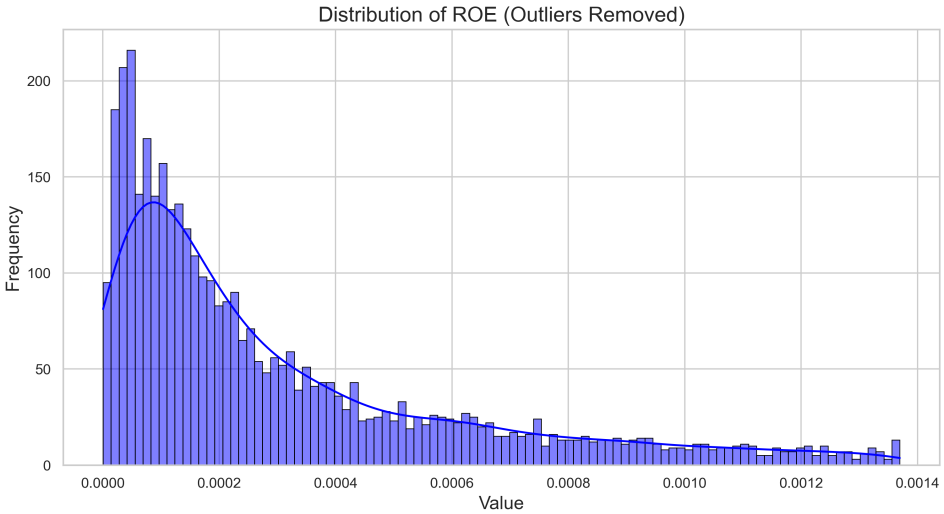
- Net Profit to Book Value:



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Example: Impact of Outliers on XS Distribution

- Net Profit to Book Value with outliers removed by MAD:



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Code Example

2 Data Preprocessing

```
# remove blacklist, untradable stocks, outliers, and stocks not traded in SH/SZ

def get_clean_descriptor(descriptor):
    # drop blacklist
    descriptor_drop_blacklist = descriptor.loc[lambda s: ~s.index.isin(blacklist.
        index)].dropna()

    # remove outliers
    descriptor_drop_outlier = descriptor_drop_blacklist.groupby(level='date',
        group_keys=False).apply(winsorize)

    # drop untradable
    descriptor_drop_untradable = descriptor_drop_outlier.loc[lambda s: ~s.index.
        isin(untradable.index)].dropna()

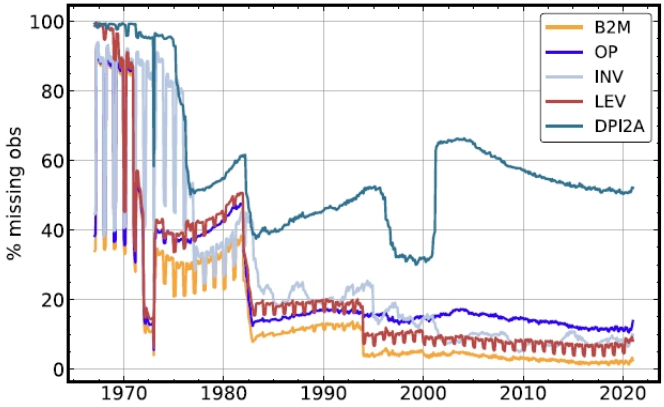
    # keep only SH and SZ exchanges
    descriptor_keep_SHSZ = descriptor_drop_untradable.loc[lambda s: s.index.isin(
        data_index)].dropna()

    return descriptor_keep_SHSZ
```

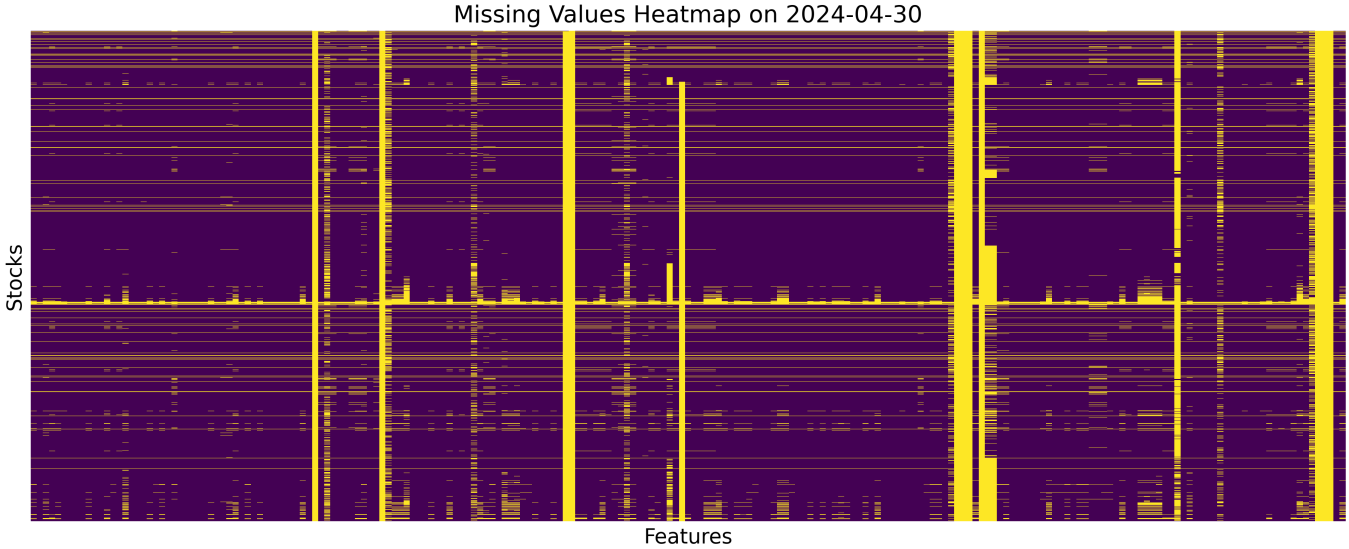

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- Introduction:
  - Missing data is a common issue in quantitative investment due to incomplete data collection, reporting delays, or data errors.
  - Solutions range from simple heuristics to advanced techniques.

- Two Main Approaches:
  - **Removal:** Costly as it reduces sample size and may introduce bias.
  - **Imputation:** Preferred but relies on assumptions. The choice of imputation method depends on the data type and context.



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Missing Data – Imputation

- Basic Imputation:
  - Use median/mean of past values for the stock. May also consider using rolling windows (e.g., 12-month rolling median) to capture recent trends without being forward-looking.

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- Time-Series Imputation:
  - Replace missing x_t with x_{t-1}. This may work only when the data is highly persistent.

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- Time-Series Imputation:
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- Cross-Sectional Imputation:
  - Use median/mean across assets for the same time period. May also use industry-specific medians to avoid distorting the distribution.

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Missing Data – Imputation

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- Cross-Sectional Imputation:
  - Use median/mean across assets for the same time period. May also use industry-specific medians to avoid distorting the distribution.
- Advanced Techniques:
  - Use latent factor models to capture both time-series and cross-sectional dependency of data to impute missing values and allows for general systematic patterns of missingness (Bryzgalova et al. 2022).

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Missing Data – Industry Practice

- Use Domain Knowledge: Leverage financial theory and industry-specific knowledge to guide imputation (e.g., impute missing credit ratings using ESG ratings).
- Robustness Checks:
  - Perform sensitivity analysis to assess the impact of imputation on model results.
  - Test alternative imputation methods to ensure robustness.
- Collaborate with Data Providers: Work with data vendors to resolve missing data issues at the source.
- Machine Learning Pipelines:
  - Integrate imputation into end-to-end ML pipelines.
  - Monitor imputation performance in production systems.

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Missing Data – Returns

- Listed companies may abuse the suspension system, resulting in intermittent trading of their stocks, which disrupts the calculation of certain metrics.

> For instance, when trying to calculate volatility using the return data over the past month, if a stock has been suspended on most trading days and only a few samples are available for the calculation, the result may be affected by the small sample size, leading to bias.

- A common practice is that the calculation should only be performed when **the number of valid samples exceeds two-thirds of the time window**.

> Taking monthly volatility as an example, since a month typically consists of 21 trading days, volatility will only be calculated when the sample size is 14 or more; otherwise, the result will be treated as a missing value.

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Industry Practice 1: Standardization

- A common practice in the industry is to normalize raw covariates across the cross-section by demeaning and standardizing them into z-scores:

$$z_i = \frac{x_i - \bar{x}}{\sigma_x},$$

where:

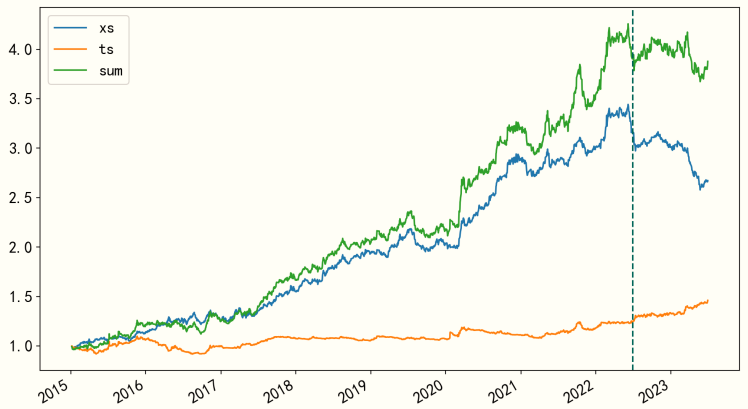
- x_i is the raw covariate of stock i,
- $\bar{x}$ is the cross-sectional mean of the covariate,
- σ_x is the cross-sectional standard deviation.

Demeaning may discard valuable information contained in the cross-sectional mean of the covariate, potentially reducing predictive power.

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- Portfolio return decomposition:

$$
\begin{aligned}
\text{Return} &= \sum_{t=1}^{T}\sum_{i=1}^{n} x_{t-1,i} \times r_{t,i} \\
&= \sum_{t=1}^{T}\sum_{i=1}^{n} (x_{t-1,i} - \bar{x}_{t-1}) \times r_{t,i} \\
&\quad + \sum_{t=1}^{T} \bar{x}_{t-1} \times \left(\sum_{i=1}^{n} r_{t,i} \right).
\end{aligned}
$$



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Industry Practice 2: Neutralization

- It is also common to neutralize raw predictors by performing a cross-sectional regression of raw predictor on the log of market capitalization and industry dummy variables:

$$x_i = \alpha + \beta_1 \ln(\text{MCap}_i) + \sum_{j=1}^{I-1} \gamma_j \cdot D_{ij} + \epsilon_i,$$

  where:
  - $\ln(\text{MCap}_i)$ is the logarithm of market capitalization for stock i,
  - D_{ij} is the dummy variable for industry j (it takes value 1 if stock i belongs to industry j, otherwise 0),
  - ϵ_i is the error term.

- The residual $\hat{\epsilon}_i$ represents the neutralized predictor. (But is industry practice always a golden rule?)

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Comments

- In general, reliable data vendors provide data that meets point-in-time requirements.
- Moreover, data vendors often construct derivative variables, which greatly facilitates factor research.
- However, as researchers, we have the responsibility to understand how the data vendors handle and process the data.
- Additionally, being familiar with the distribution of the variables you are using is crucial for selecting the appropriate preprocessing methods.

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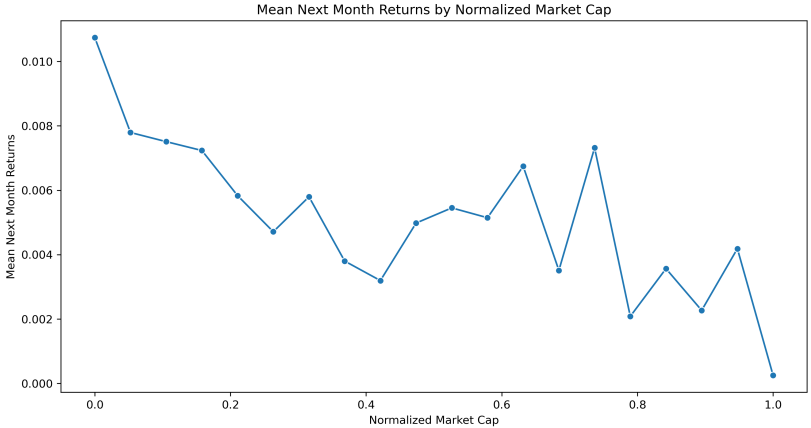
3 Feature Selection and Labeling

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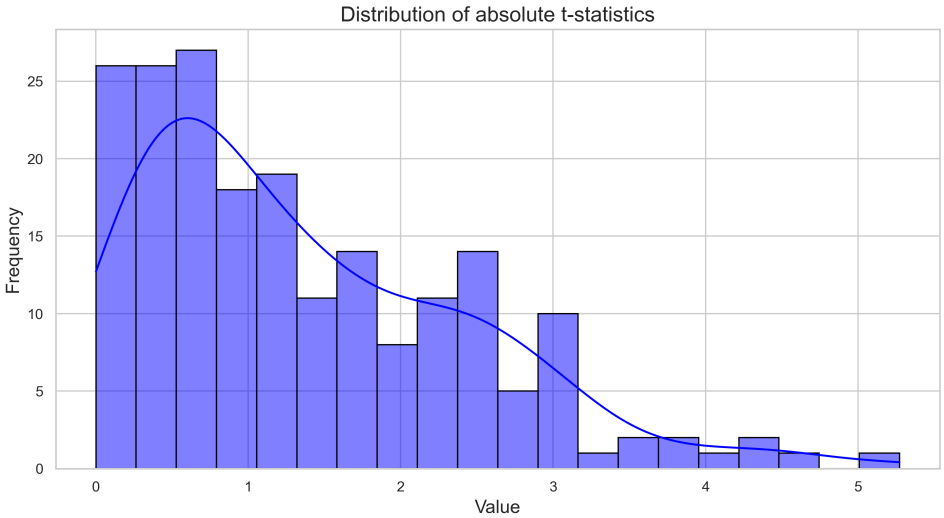
- Portfolio sort can be the first step. (Or you can calculate information coefficient.)

- Theoretical underpinning:
  - Suppose there is only one feature X and that we seek a model $Y = f(X) + \text{error}$.
  - The choise of $f(X)$ that minimizes the mean squared error, i.e., $\mathbb{E}[(Y - f(X))^2]$, is the conditional expectation $f(X) = \mathbb{E}[Y|X]$. .



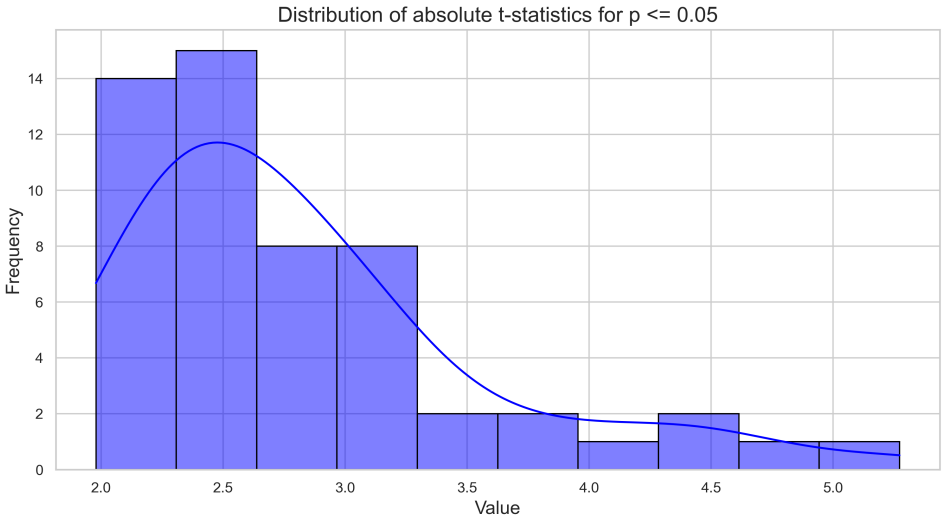
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- Not all covariates carry predictive information.



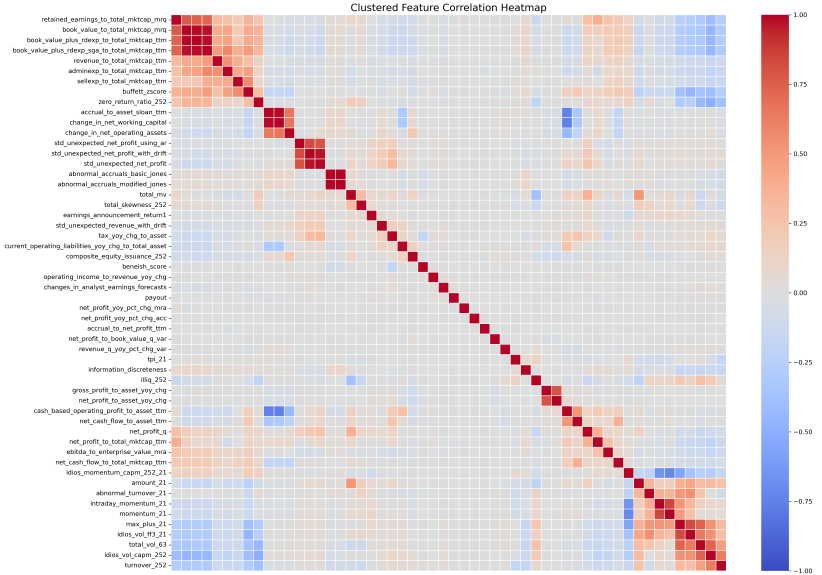
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- Filter covariates by $p-\text{value} \leq 0.05$. However, be careful about the multiple hypothesis testing issue!



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- Highly correlated covariates are not necessarily redundant because covariates are noisy proxies of the true return source.
- However, in linear models, high correlation may cause multicollinearity.



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Example: Multicollinearity

- *Multicollinearity* occurs when two or more independent variables are highly correlated.
- The amount of multicollinearity can be measured by variance inflation factor (VIF), which is defined for x_i as:

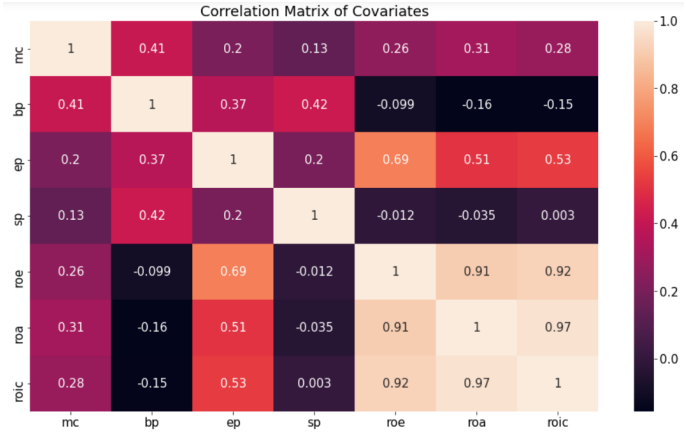
$$\text{VIF} = \frac{1}{1 - R_i^2}, \tag{1}$$

  where R_i^2 is derived from regressing x_i on all other independent variables. **A VIF between 1 and 5 indicates no severe multicollinearity issue, but a VIF higher than 5 indicates severe multicollinearity.**

- The existence of multicollinearity inflates the standard errors of regression coefficients, and this in turn increases the variance of the model, i.e., **model is sensitive to the change in input data. This is BAD for a quantitative model or strategy in practice.**

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- Consider the problem of predicting stock returns using fundamentals.
- Independent variables from two categories:
  - **Value**: book-to-market ratio (BP), earnings-to-price ratio (EP), sales-to-price ratio (SP)
  - **Profitability**: return on equity (ROE), return on total asset (ROA), return on invested capital (ROIC)
  - Besides, logarithm of market cap, i.e., ln(market cap). (This is a common treatment in practice.)



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	coef	std err	t	P>|t|	[0.025	0.975]
const	0.0211	0.005	4.114	0.000	0.011	0.031
mc	-0.0126	0.006	-1.941	0.053	-0.025	0.000
bp	0.0085	0.008	1.048	0.295	-0.007	0.024
ep	-0.0412	0.010	-4.076	0.000	-0.061	-0.021
sp	0.0045	0.006	0.777	0.437	-0.007	0.016
roe	0.0439	0.019	2.344	0.019	0.007	0.081
roa	0.0435	0.023	1.914	0.056	-0.001	0.088
roic	-0.0423	0.025	-1.680	0.093	-0.092	0.007

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- All three profitability category variables, i.e., ROE, ROA and ROIC, have large VIFs.
- For example, the VIF of ROIC is over 24. These numbers indicate severe multicollinearity among the three variables.
- For the rest of the independent variables, multicollinearity is not a (big) concern.

	VIF
mc	1.598184
bp	2.477554
ep	3.892371
sp	1.270993
roe	13.349351
roa	19.607751
roic	24.127641

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- OLS coefficient (-0.0423) suggests that ROIC is **negatively** correlated to next month returns. This means that, all else being equal, firms with lower profitability measured by ROIC will earn higher expected returns.
- Does this make sense to you? Given that profitability is a well known equity factor (Fama and French 2015), we should expect just the opposite.
- As a matter of fact, the correlation coefficient between ROIC and next month return is 0.117. So, ROIC and return are **positively** correlated.

What went wrong???

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- Multicollinearity is the key. Since ROIC is highly and positively correlated with ROE and ROA, in OLS, the negative regression coefficient is required to **offset** the positive regression coefficients of ROA and ROE.
- OLS results: their regression coefficients have similar absolute values, and differs only in their signs.
- Regularization (i.e. penalized regression) or dimensionality reduction can help address this issue. (We will talk more in future lectures.)

	Coefficient
ROE	0.0439
ROA	0.0435
ROIC	−0.0423

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Incorporating Macro-Economic Variables

- Motivation: Financial data should not be analyzed in isolation; it is influenced by its environment, often proxied by macro-economic indicators.

- Feature Augmentation: Multiply the original feature $x_{i,t}$ by an exogenous macro-economic indicator z_t:

$$\hat{x}_{i,t} = z_t \times x_{i,t}.$$

  Example: Gu, Kelly and Xiu (2020) use 8 economic indicators, expanding the feature space ninefold.

- Benefits:
  - Captures the impact of macro-economic conditions on financial variables.
  - Integrates external economic trends into the model.

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Labeling

- Labeling refers to the process of defining target variables (labels). Choosing the right label is crucial for model performance and portfolio outcomes.

- Common types of labels include (Coqueret and Guida 2023):
  - Raw Asset Returns: Direct use of future asset returns as labels.
  - Future Relative Returns: Returns relative to a benchmark (e.g., market index or sector portfolio).
  - Probability of Positive Return: Likelihood of returns exceeding a specified threshold.
  - Probability of Outperforming Benchmark: Likelihood of outperforming a benchmark over a given time frame.
  - Binary Labels: YES (outperforming) vs. NO (underperforming).
  - Risk-Adjusted Labels: Sharpe ratios, information ratios, etc.

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4 Machine Learning Concepts

- Financial Data
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- Consider the essential problem in quantitative investment:

$$R^e_{i,t+1} = \mathbb{E}[R^e_{i,t+1}|\boldsymbol{x}_{i,t}] + e_{i,t+1}, \quad \mathbb{E}[R^e_{i,t+1}|\boldsymbol{x}_{i,t}] = f(\boldsymbol{x}_{i,t}).$$

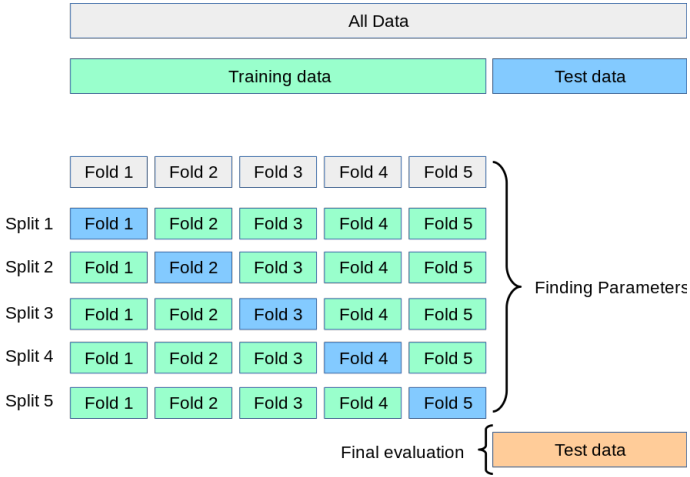
- Select a class of models (e.g., neural networks) $\hat{f} \in \mathcal{F}$, and learn its parameters from the data under a given loss function L. To guard against overfitting, an important concept is **regularization**. Therefore:

$$\min \frac{1}{n}\sum_{i=1}^{n} L(\hat{f}(\boldsymbol{x}_i), y_i) + S(\hat{f}).$$

- Regularization procedures rely on the choice of hyperparameters. These hyperparameters are tuned via validation.

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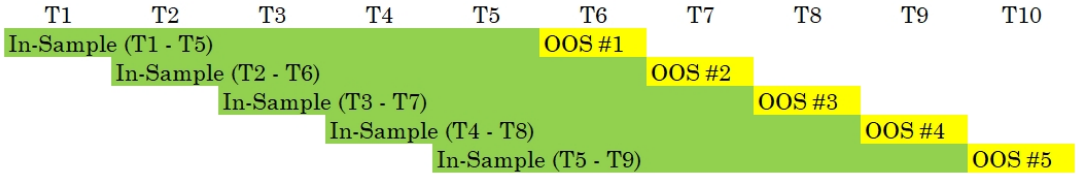
- Steps:
  - Split data into k smaller subsets called *folds*.
  - Train the model on $k-1$ folds and validate it on the remaining fold.
  - Repeat this process k times, with each fold serving as the validation set once.
  - Compute the average performance across all k test folds.
- **The sequential nature of data is not explicitly accounted for.**



Source: https://scikit-learn.org/stable/modules/cross_validation.html

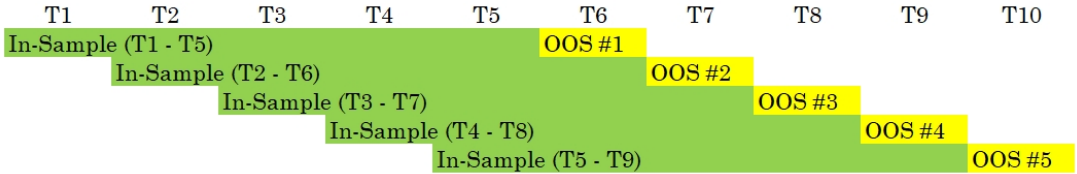
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- Rolling WF:

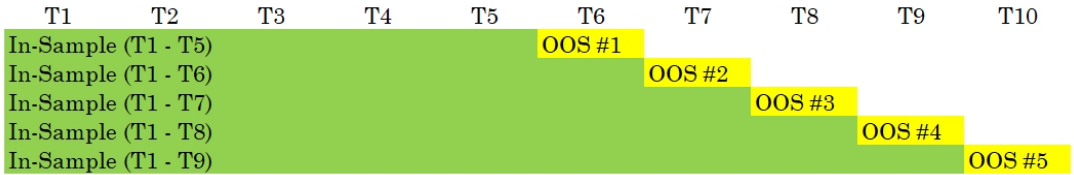


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- Rolling WF:



- Expanding or Anchored WF:



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Combinatorial Purged Cross-Validation (CPCV)

- Motivation: In WF and CV, each data point is only used in a **single test set**.
- CPCV generates **multiple backtesting paths**, enabling robust performance evaluation across various market scenarios.
- Suppose data is split into N subsets and K of them are used for testing. Then, number of different paths is:

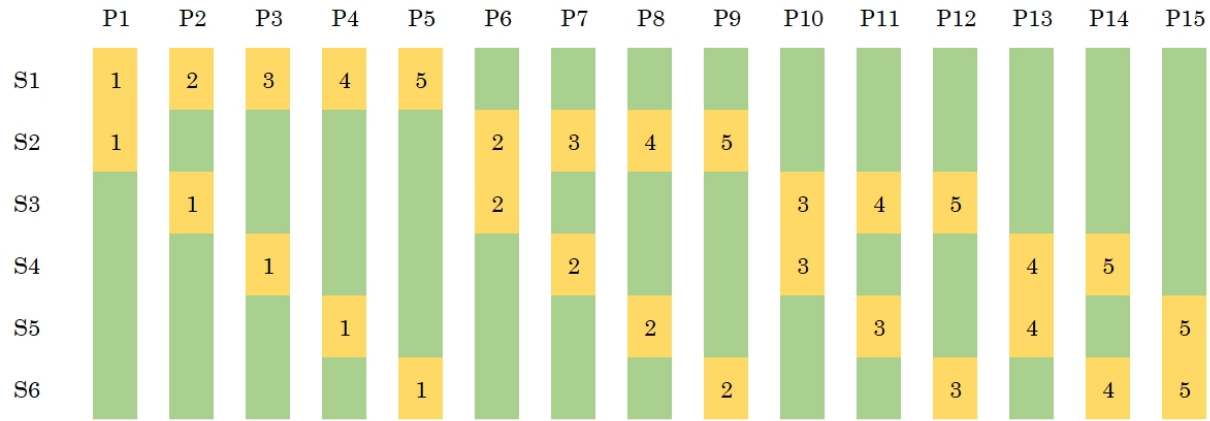
$$\left(\begin{array}{c} N \\ K \end{array} \right).$$

- Combines combinatorial splits with **purging** to address data leakage and dependency issues.

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Multiple Backtesting Paths in CPCV

- Assume $N = 6$ and $K = 2$.



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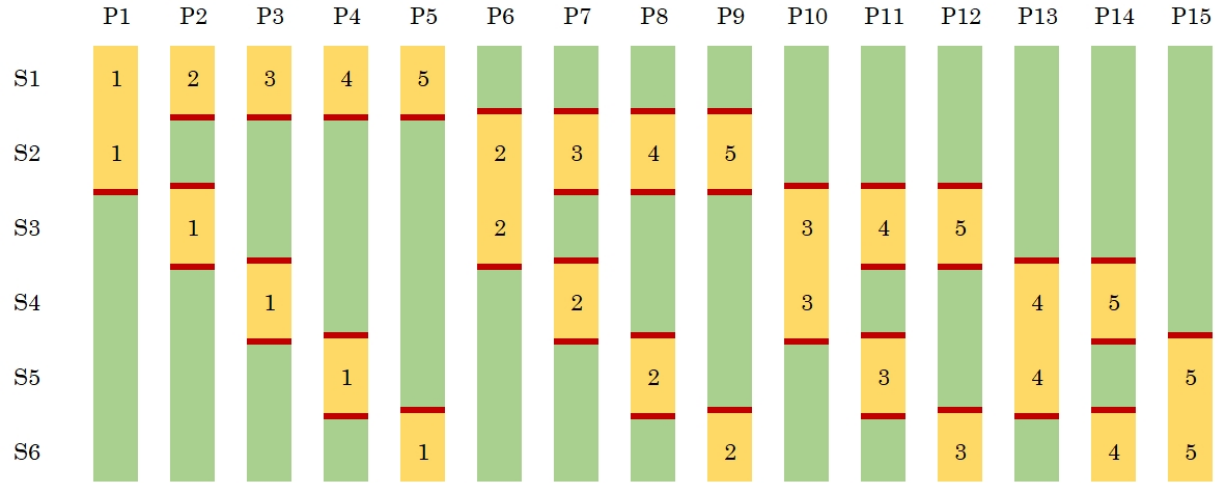
Data Leakage in Time Series

- **The Problem:** Sequential data points (e.g., x_{t-1}, x_t) are often highly correlated.
- **How It Happens:**
  - Training set includes x_{t-1}, test set includes x_t.
  - Model indirectly learns x_t from x_{t-1}, violating the unseen data principle.
- **Impact:** Overestimates performance; fails to reflect real-world scenarios.

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Purging in CPCV

- Purging: Introduce a **buffer** between training and test sets.



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- Steps:
  1. Divide the dataset into K subsets (folds).
  2. Select combinations of folds for training and testing.
  3. Purge adjacent periods between training and testing on the training side.
  4. Use all valid paths to train and evaluate the model.

- Outcome:
  - Comprehensive evaluation across multiple paths.
  - Robust estimates of out-of-sample performance.

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CPCV vs. WF and CV

- Advantages:
  - Generates multiple paths for robust evaluation.
  - Mitigates data leakage with purging.
  - Provides a more reliable measure of strategy performance.

- Comparison with Traditional Methods:
  - **k-Fold Cross-Validation**:
    - Assumes i.i.d. data.
    - Fails to account for sequential dependencies.
  - **Walk-Forward Validation**:
    - Accounts for time order but produces only one backtesting path.
  - **CPCV**:
    - Combines the best of both worlds.
    - Creates multiple paths for robust validation.

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- R-squared (similar to MSE):

$$\text{R-squared} = 1 - \frac{\sum_{(i,t)\in\mathcal{T}_{\text{val}}}(r_{i,t+1}-\hat{r}_{i,t+1})^2}{\sum_{(i,t)\in\mathcal{T}_{\text{val}}} r_{i,t+1}^2},$$

  where $\mathcal{T}_{\text{val}}$ indicates validation sample.

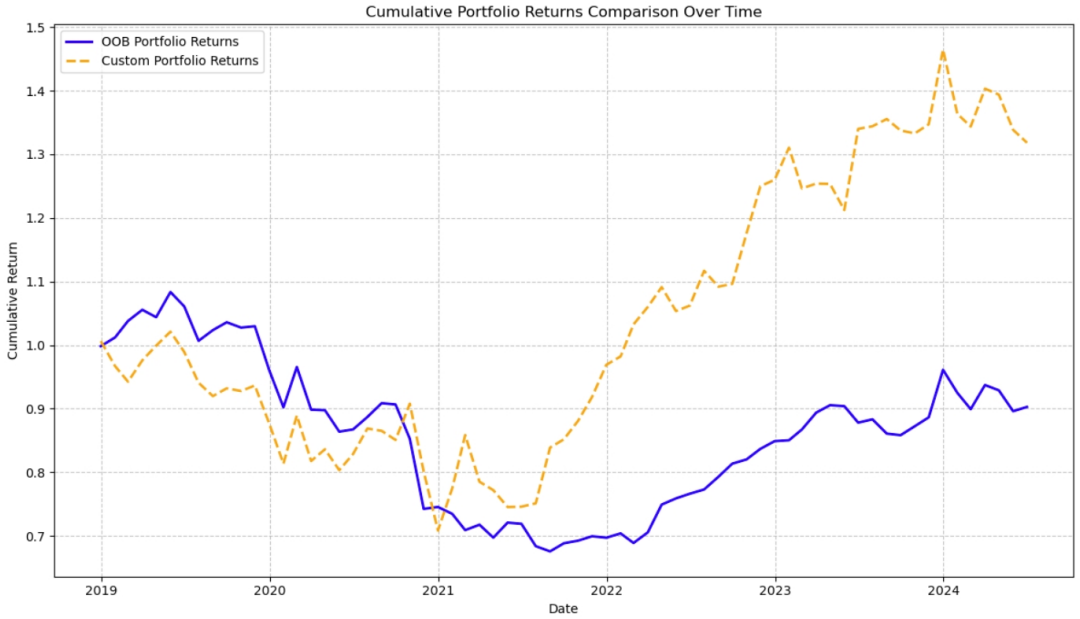
- Information Correlation (IC):

$$\text{IC} = \text{corr}(r_{i,t+1}, \hat{r}_{i,t+1}).$$

- Sharpe Ratio:
  - Form a portfolio with weights $w_{i,t+1} = \hat{r}_{i,t+1} / \sum_j |\hat{r}_{j,t+1}|$.
  - Evaluate the Sharpe ratio of this portfolio on the validation set.

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Hyperparameter Tuning Criteria – The Choice Matters!



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5 Facts and Fictions

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Fiction 1: Data will "Speak for Itself"

- Fallacy of Data Speaking for Itself:
  - Feeding raw financial data into ML algorithms without theoretical guidance often fails.
  - Financial data has low signal-to-noise ratio and non-stationarity, making out-of-the-box ML ineffective.

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- Critical Considerations in Asset Pricing:
  - Prior distributions, scaling of covariates, regularization penalties, and tuning criteria significantly impact outcomes.
  - Example: Recall Nagel's example!

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- Critical Considerations in Asset Pricing:
  - Prior distributions, scaling of covariates, regularization penalties, and tuning criteria significantly impact outcomes.
  - Example: Recall Nagel's example!

- Exhaustive permutation testing is infeasible (and I don't recommend it!); leverage financial theory for priors.

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Fact 1: Apply Machine Learning within Asset Pricing Framework

- Integration of Machine Learning and Asset Pricing:
  - Leverage big data and machine learning within the framework of asset pricing theory.
  - Evolution from CAPM to APT/ICAPM and factor models has driven empirical progress.

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- Common Themes:
  - Use of extensive covariates beyond traditional research.
  - Focus on out-of-sample (OOS) portfolio performance.
  - Modeling factor exposures (β) as functions of covariates (firm characteristics, macroeconomic variables).

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- Unified Framework:
  - Despite different machine learning algorithms, all approaches align with SDF and multifactor models.
  - Demonstrates the synergy between advanced machine learning techniques and asset pricing theories.

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Fiction 2: Machine Learning Models Neglect Interpretability

- Misconception: Machine learning models are often perceived as black boxes, leading to the belief that academic research neglects interpretability.

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- Interpretability in Linear Models:
  - PCA (KNS 2018, 2020): First two principal components correspond to size and value factors.
  - IPCA (KPS 2019): Linear combination of managed portfolios through cross-sectional regression.

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- Interpretability in Nonlinear Models:
  - While less intuitive, academic research emphasizes on feature importance (by, for example, permutation importance, gradients in a neural network, etc.).
  - See Fact 2!

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Fact 2: Machine Learning Unveils Key Covariates

- Identifying Predictive Variables: - Machine learning helps uncover the most important predictive variables, aligning with empirical asset pricing findings.

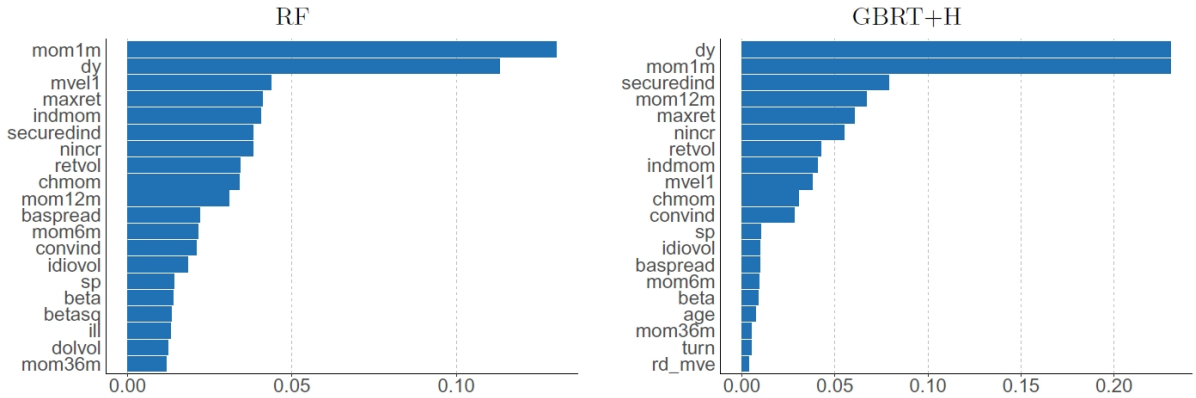
- Examples:
  - Gu, Kelly and Xiu (2020) use permutation Importance to Identify crucial covariates for expected returns, including momentum/reversal, liquidity-related, risk-related (e.g., idiosyncratic volatility), and fundamentals.

  - Chen, Pelger and Zhu (2023) calculate partial derivatives of SDF weights to assess model interpretability.

  - Avramov, Cheng and Metzker (2023) observe selected stocks across covariates to infer variable significance.

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Fact 2: Machine Learning Unveils Key Covariates



- Source: Gu, Kelly and Xiu (2020)

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Fiction 3: Complex Models Lead to Poor OOS Performance

- Traditional View:
  - Complex models may overfit in-sample data, leading to poor OOS performance.
  - Bias-Variance Trade-off: Low complexity leads to underfitting (high bias, low variance); high complexity leads to overfitting (low bias, high variance).

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  - Complex models may overfit in-sample data, leading to poor OOS performance.
  - Bias-Variance Trade-off: Low complexity leads to underfitting (high bias, low variance); high complexity leads to overfitting (low bias, high variance).
- Latest Development:
  - "Double Descent" (Loog et al. 2020; Hastie et al. 2022): Increasing parameters beyond training data size can improve OOS performance.
  - Empirical Evidence: Over-parameterized deep learning models can generalize well despite high complexity.

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Fact 3: With Regularization, Complexity > Cost

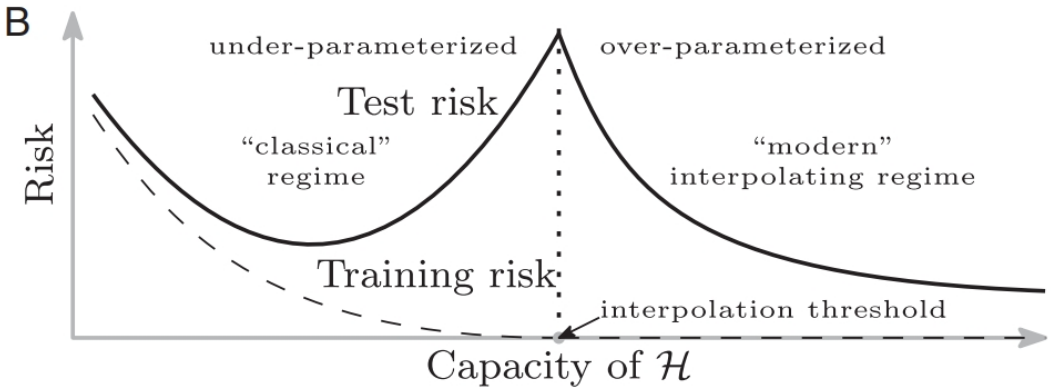
The true data-generating process (DGP) is unknown; all models are approximations of an unknown and potentially misspecified DGP.

Model	Simple			Complex
True	$K = 1000$	$K = 1000$	$\cdots$	$K = 1000$
Empirical	$K = 1$ (CAPM)	$K = 3$ (FF3)	$\cdots$	$K = 1000$

- Adding more variables comes with substantial statistical costs (over-parameterization).
- Compared to simple models, complex models provide a better approximation of the true DGP.
- Empirical results: better approximation of the DGP outweighs the statistical cost of adding parameters (Didisheim et al. 2023; Kelly, Malamud and Zhou 2024).

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"Double Descent"



- Source: Loog et al. (2020)

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Fiction 4: Nonlinear Models Can Double SR Easily

- Misconception: Nonlinear models are often believed to easily double the Sharpe ratio compared to traditional linear factor models.

- Empirical Results (Baba-Yara, Boyer and Davis 2021):
  - Machine learning models show higher Sharpe ratios, but traditional models use fewer covariates, making comparisons unfair.
  - PCA related models (e.g. IPCA) achieves higher OOS Sharpe ratios than some nonlinear models (random forests, neural networks).

Machine Learning Models	OOS Sharpe Ratio
BPZ Random Forest	2.19
Davis Neural Network	3.21
KNS PCA	3.21
KPS IPCA	3.39

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Fact 4: Nonlinear Models Offer Marginal Gains

- Nonlinear Relationships: Nonlinear relationships exist between covariates and asset returns.
- Interaction Terms:
  - Crucial in nonlinear models but impractical in traditional linear regression with many variables.
  - E.g., GAN of CPZ(2023) captures interactions but individual covariates' impact on SDF is nearly linear.
- Robustness and Tuning:
  - Fluctuations in Results: Significant across different empirical periods.
  - Research Protocol Needed: Lack of discussion on tuning and robustness under different hyperparameters.

Machine Learning Models	OOS Sharpe Ratio
BPZ Random Forest	0.86 (2.19)
Davis Neural Network	2.39 (3.21)
KNS PCA	2.77 (3.21)
KPS IPCA	3.21 (3.39)

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- Bryan Kelly (Yale and Head of Machine Learning at AQR) on machine learning:

> **Linear models are first order approximations. First order approximations are the most important approximations. When you try to set people's expectations of what you're going to get out of macine learning, you shouldn't expect rvolutionary performance from machine learning. You should expect second and third order of improvements, because the first order models we are use are good. They leave some predictability on the table because there are those non-linearities that we can exploit. But you should not expect it to double or triple your Sharpe ratio. Seeing something like a 20 percent increase in Sharpe Ratio from introducing non-linearities feels realistic to me.**

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Fiction 5: ML Models Can Be Easily Applied in Practice

- Misconception: Machine learning models are often perceived as easily applicable in practice, but practical challenges exist.
- High Turnover Ratios:
  - Empirical results from Avramov, Cheng and Metzker (2023):

Machine Learning Models	Monthly Turnover Ratio
Gu, Kelly and Xiu (2020)	0.976
Chen, Pelger and Zhu (2023)	1.664
Kelly, Pruitt and Su (2019)	1.186
Gu, Kelly and Xiu (2021)	1.565

  - Traditional factors (e.g., size, value) have turnover rates below 10%. High turnover rates make it challenging to achieve excess net returns after costs.
- Jensen et al. (2022) integrated transaction costs into portfolio optimization.

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Fact 5: Limited Value for Some Institutional Investors

- Machine learning models uncover predictability in stocks with high arbitrage and transaction costs, limiting their value for some institutional investors.

- Empirical Findings:
  - Excess returns often come from short legs or tiny market cap stocks.
  - Anomalies are concentrated in small-cap stocks.
  - Significant performance decline in nonlinear models in subsamples (excluding tiny caps, unrated firms, distressed firms).
  - Linear models (e.g. IPCA) produce consistent performance across full sample and subsamples.

- Successfully adapting nonlinear models to focus on stocks with lower arbitrage and transaction costs is crucial for practical implementation.

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Model	Full Sample	Exclude Tiny Market Cap	Exclude No Credit Rating	Exclude Financially Distressed
Panel A: Absolute Return (%)				
Gu, Kelly and Xiu (2020)	1.56	1.05	1.02	0.72
	(4.53)	(3.24)	(3.18)	(2.49)
Chen, Pelger and Zhu (2023)	2.13	1.08	0.82	0.92
	(6.37)	(4.06)	(2.83)	(2.91)
Kelly, Pruitt and Su (2019)	0.95	0.91	0.89	0.73
	(5.62)	(5.57)	(5.08)	(4.08)
Gu, Kelly and Xiu (2021)	1.16	1.11	0.87	0.67
	(4.17)	(4.22)	(2.97)	(2.22)
Panel B: Alpha (%) against FF5 + MOM				
Gu, Kelly and Xiu (2020)	0.92	0.31	0.43	0.20
	(4.08)	(1.51)	(2.05)	(0.92)
Chen, Pelger and Zhu (2023)	1.87	0.55	0.42	0.57
	(4.86)	(2.23)	(1.46)	(1.82)
Kelly, Pruitt and Su (2019)	0.62	0.61	0.61	0.43
	(3.31)	(3.71)	(3.38)	(2.37)
Gu, Kelly and Xiu (2021)	0.75	0.39	0.19	0.05
	(3.01)	(2.03)	(0.79)	(0.20)

Data excerpted from Avramov, Cheng and Metzker (2023).

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Questions?

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